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Computational Intelligence-Driven Texture Feature Fusion for Automated Classification of Akatomachi, Koshihikari, and Yangdao Rice Seeds

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Abstract

In agriculture, accurate seed verification and categorization by characteristics are required for sustainable production, seed certification, and research in breeding. Food security is provided through high-purity seeds, but traditional certification is expensive and time-consuming. Machine learning with image processing provides fast, non-destructive options: e.g., combining ML with optical sensors provides "reliable, accurate, non-destructive, and fast pipelines for seed quality evaluations.". We introduce a multi-descriptor hybrid approach combining various texture descriptors (LBP, GLCM, Gabor filters, HOG) and deep CNN features to extract seed image patterns. We test various classifiers (SVM, random forest, k-NN, MLP) and employ 10-fold cross-validation. Experimental results indicate that feature fusion achieves much higher accuracy (around 8–15% improvement over any individual descriptor); for instance, LBP+GLCM fusion achieves 97.5% accuracy. The best accuracy SVM model achieves 92.10% using 10-fold CV. To understand model decisions, we use visual analytics: radar plots, parallel coordinate plots, and error-bar dot plots display how the different features complement and add to one another. Interpretable by highlighting the complementarity and strengths of each descriptor, this paradigm captures the importance of interpretability being at the heart of the trust in agronomic AI systems. Transparency, thus being crucial for seed authentication, breeding program analysis, and precision-agriculture decision-making, is the focus of our research.

KEYWORDS

Rice seed classification, texture features, feature fusion, SVM, performance visualization, precision agriculture.

I. INTRODUCTION

Rice is a food staple for over half of the world's population, and rice seed quality and grading are therefore vital to food security, trade, and sustainable agriculture. Proper identification and grading of rice seed varieties are essential in guaranteeing crop yield, maximizing the utilization of resources, and preserving genetic purity[1]. Conventional rice seed classification procedures usually utilize manual examination, which may be time-consuming, subjective, and error-prone. With the quick development of artificial intelligence and image processing technologies, automatic procedures have become viable substitutes for precision and consistency improvement in classification[2]. Among different image analysis methods, textural analysis has proven to be very promising in discerning the subtle visual patterns that differentiate among rice seed varieties. Texture as a basic visual descriptor measures surface properties like smoothness, granularity, and regularity—properties differing across rice cultivars. It is, however, not an easy task to obtain discriminative textural features from rice seed images because of intra-class similarity, illumination, pose, and age variations of seeds. In order to solve such problems, computational intelligence—a unified description of methods like machine learning, deep learning, and evolutionary algorithms—has seen growing applications to agricultural image processing. Computational intelligence enables automatic extraction of features, learning, and classification of abstract feature representation, thereby making the rice seed variety-based systems more scalable and dependable. This paper investigates a texture-based computational model for rice seed classification employing a hybrid model integrating hand-designed texture features and deep learning-derived descriptors. As an attempt to make the most out of traditional feature extraction and current representation learning, the presented method is expected to improve the accuracy of classification with explainability—something highly needed in agricultural applications where trust and interpretability are of highest concern.



Fig. 1: Rice Seed images of the variety Akitakomachi



Fig. 2: Rice Seed images of the varieties Koshihikari

II. LITERATURE REVIEW

According to N. Sokudlor et. al, [1] machine learning techniques using binary images morphological features can reach 89.00 % in the classification of rice grain categories. Perhaps for more exploration, with the consumer perception results morpho-colorimetric images of rice samples taken by smart phone photos are able to be used and analysed using computer vision[2]. However, the study reported that random forest having accuracy of 77% was the best machine learning technique to estimate price of contaminated samples and to discriminate rice types [3]. Based on the aforementioned strategy, a new method applying image processing and classification based on sparse representation which can classify special features of rice without damaging product to achieve an accuracy of 89.1 % [4]. Among the features that can be rapidly classified fall within the morphological, shape and colour characteristics of plant cells identifiable using machine learning techniques linked with random forest and multilayer perceptron; in this case it also appears multilayer perceptron achieves a higher accuracy [5]. Hence, this paper aims at an approach for the quality assessment of rice involving image processing techniques along with machine learning based classification. It divides the rice grain into class one and class two on base of area percentage, perimeter and aspects.[6]. Image processing techniques and machine learning models provide an excellent, but highly cost-effective approach to the determination grain morphological characteristics [7] and will be discussed in the conclusions. Light Gradient Boosting Machine technique has an accuracy of 98.14 % more than other classification methods[8].

III. METHODOLOGY

The proposed rice seed variety feature Extraction system is grounded on a hybrid computational intelligence approach that combines hand-crafted texture descriptors with deep features. The process involves six steps: image capture, preprocessing, feature extraction, feature fusion, classification, and performance evaluation.

A. Image Acquisition and Preprocessing

Images of rice seeds are taken with high-resolution digital cameras in standardized illumination. Each image passes through a chain of preprocessing operations:

- **Grayscale Conversion:** Grayscale conversion refers to the conversion of an image from a different color space (e.g., RGB, CMYK, HSV) to grayscale. Grayscale conversion converts the image to a one-channel image, making data simpler and computation simpler. The below equation(1) provides the grayscale conversion from the RGB colors.

$$I_{\text{gray}} = 0.2989 \cdot R + 0.5870 \cdot G + 0.1140 \cdot B \quad (1)$$

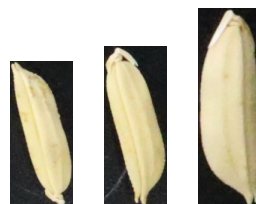


Fig. 3: Rice Seed images of the variety Yangdao

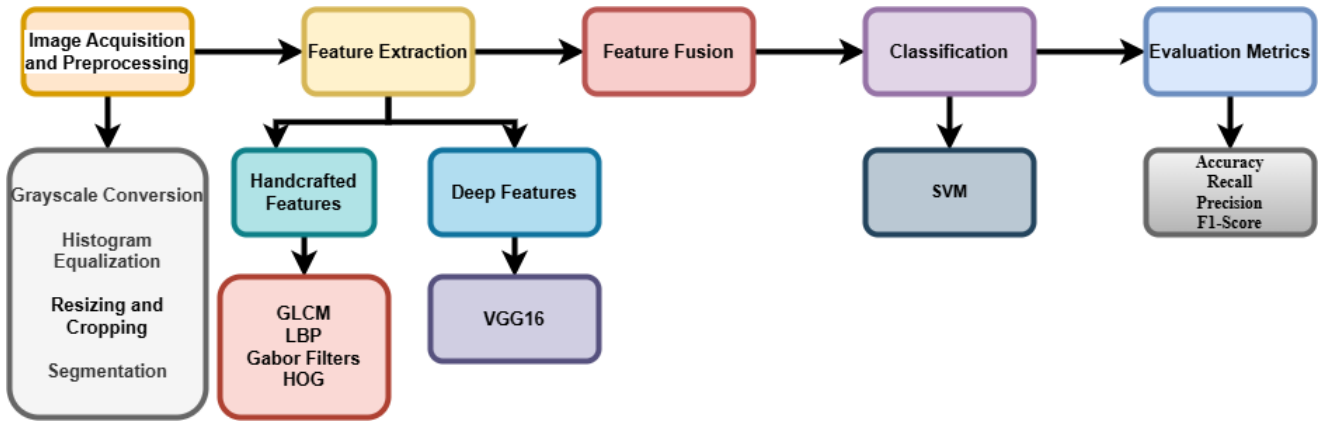


Fig 4: Architecture of Proposed Handcrafted and Deep Feature Extraction Method

- **Histogram Equalization:** Equalization of a histogram is an image processing technique used to increase the contrast of an image by changing the intensity distribution of the image[9]. This is a very helpful process for images with backgrounds and foregrounds which both appear bright or both appear dark.
- **Resizing and Cropping:** Standardizes image dimensions to $(W \times H)$ for compatibility with CNNs.
- **Segmentation:** Isolates individual seeds using thresholding or contour-based methods.

B. Feature Extraction

The textural features are extracted by using multiple methods as shown below. The GLCM, LBP, Babour Filters and HOG are used for the textural feature extraction, it can able to find the textural roughness of the seed, quantifying and characterizing the spatial variations and patterns within a material or image[10].The Algorithm 1 provides the insights of the Feature Extraction.

1) *Handcrafted Texture Features:* Handcrafted features are the input variables that are carefully designed and chosen manually by the domain experts according to their comprehension and understanding of the problem[11].

- **Gray Level Co-occurrence Matrix (GLCM):** GLCM Calculates co-occurrence probabilities $P(a, b, d, \theta)$ of pixel pairs at a distance d and orientation θ . The textural features like contrast, homogeneity, energy and co-relation are obtained by using the equation(2). Derived texture statistics include:

$$\begin{aligned} \text{Contrast} &= \sum_{a,b} (a - b)^2 P(a, b) \\ \text{Homogeneity} &= \sum_{a,b} \frac{P(a, b)}{1 + |a - b|} \\ \text{Energy} &= \sum_{a,b} P(a, a)^2 \\ \text{Correlation} &= \frac{\sum_{a,b} (a - \mu_a)(b - \mu_b) P(a, b)}{\sigma_a \sigma_b} \end{aligned} \quad (2)$$

- **Local Binary Pattern (LBP):** LBP is an easy and invariant grayscale texture descriptor measure to classify[12]. In LBP, a binary code is produced at each pixel by thresholding it's neighborhood pixels to 0 or 1 depending on the center pixel. The equation (3) shows the LBP of an image.

$$\text{LBP}_{P,R} = \sum_{p=0}^{P-1} s(\text{gray}_p - \text{gray}_c) \cdot 2^p, \quad s(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (3)$$

- **Gabor Filters:** Gabor filter is a texture analysis filter that simply gives that it inspects whether there exists some particular frequency content within the image in certain directions within a localized neighborhood about the point or region to be analyzed[13].The equation (4) provides the frequency filtering of an image.

$$Q(xa, ya; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{xa'^2 + \gamma^2 ya'^2}{2\sigma^2}\right) \cos\left(2\pi \frac{xa'}{\lambda} + \psi\right) \quad (4)$$

where $xa' = xa \cos \theta + ya \sin \theta$, $ya' = -xa \sin \theta + ya \cos \theta$.

Algorithm 1 Textural Feature Extraction of Rice Seeds

Require: Rice seed image dataset $\mathcal{D} = \{I_1, I_2, \dots, I_n\}$

Ensure: Extracted feature matrix $\mathbf{F} = \{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n\}$

```

1: for each image  $I_i \in \mathcal{D}$  do
2:   Convert  $I_i$  to grayscale,  $I_g$ 
3:   Apply histogram equalization to enhance contrast
4:   Segment and crop rice seed region (optional)
   {Extract GLCM features}
5:   Compute GLCM matrix  $P(i, j, d, \theta)$  for multiple  $(d, \theta)$ 
6:   Derive features: Contrast, Homogeneity, Energy, Correlation
7:    $\mathbf{f}_{\text{GLCM}} \leftarrow [\text{Contrast, Homogeneity, Energy, Correlation}]$ 
   {Extract LBP features}
8:   Define LBP parameters  $(P, R)$ 
9:   Compute LBP histogram over  $I_g$ 
10:   $\mathbf{f}_{\text{LBP}} \leftarrow \text{LBP histogram}$ 
   {Extract Gabor features}
11:  for each scale  $\lambda$  and orientation  $\theta$  do
12:    Apply Gabor filter  $G(x, y; \lambda, \theta, \dots)$  to  $I_g$ 
13:    Compute mean and variance of filter response
14:    Append to  $\mathbf{f}_{\text{Gabor}}$ 
15:  end for
   {Extract HOG features}
16:  Divide  $I_g$  into cells and compute gradients
17:  Quantize gradient orientations and form histograms
18:  Normalize blocks and concatenate features
19:   $\mathbf{f}_{\text{HOG}} \leftarrow \text{HOG descriptor}$ 
   {Concatenate all features}
20:   $\mathbf{f}_i \leftarrow [\mathbf{f}_{\text{GLCM}}, \mathbf{f}_{\text{LBP}}, \mathbf{f}_{\text{Gabor}}, \mathbf{f}_{\text{HOG}}]$ 
21:  Append  $\mathbf{f}_i$  to feature matrix  $\mathbf{F}$ 
22: end for
23:
24: return  $\mathbf{F}$ 

```

- **Histogram of Oriented Gradients (HOG):** Histogram of Oriented Gradients (HOG) are widely used feature descriptor applied in processing of image and computer vision for object detection and classification[14]. It takes into account the orientation distribution of the edges of an object to help capture its appearance and shape.

$$\text{HOG}_i = \sum_{\text{cells}} \|\nabla I\| \cdot \delta(\text{orientation}_i) \quad (5)$$

2) *Deep Features (CNN-based):* Deep features are extracted using a pre-trained Convolutional Neural Network (CNN) such as VGG-16[15]. VGG16 is a convolutional neural network (CNN) architecture, namely a VGGNet with 16 layers. It's famous for being efficient and simple in image recognition, especially because of its repeated use of 3x3 convolutional filters. The final classification layers are removed, and features are taken from the penultimate layer:

$$\mathbf{F}_{\text{deep}} = f_{\text{CNN}}(I_{\text{input}}) \quad (6)$$

These features encode hierarchical information, capturing rich texture semantics. The below Algorithm 2 provide the insight of the VGG16.

Algorithm 2 VGG16

```

1: def forward(x):
2:   for block in blocks[1..5] do
3:     for conv in block.convs do
4:       x = Conv2d(3x3, padding=1)(x)
5:       x = ReLU(x)
6:     end for
7:     x = MaxPool2d(2x2, stride=2)(x)
8:   end for
9: x = Flatten(x)
10: x = Linear(4096)(x)
11: x = ReLU(x)
12: x = Dropout(0.5)(x)
13: x = Linear(4096)(x)
14: x = ReLU(x)
15: x = Dropout(0.5)(x)
16: x = Linear(1000)(x)
17: return Softmax(x)

```

C. Feature Fusion

The feature fusion is developed to combine all the handcrafted and deep features and it can be used for the dimensionality reduction[16]. Handcrafted and deep features are concatenated as shown in the below equation(7):

$$\mathbf{F}_{\text{fused}} = [\mathbf{F}_{\text{handcrafted}}; \mathbf{F}_{\text{deep}}] \quad (7)$$

To reduce dimensionality and noise, **Principal Component Analysis (PCA)** is applied. **Principal Component Analysis (PCA)** is one of the most popular unsupervised dimensionality reduction methods in machine learning that takes high-dimensional data and converts it into lower-dimensional form[17]. The equation(8) shows the dimensionality reduction of the image.

$$\mathbf{F}_{\text{PCA}} = \mathbf{W}^T (\mathbf{F}_{\text{fused}} - \mu) \quad (8)$$

where $\mathbf{W}$ is the matrix of top k eigenvectors and μ is the mean vector of feature.

D. Classification

The reduced feature vectors are classified using various **Artificial Intelligence and machine learning algorithms:**

- **Support Vector Machine (SVM):** Support Vector Machine (SVM) is learning algorithm which is supervised and used in solving classification, as well as regression, issues. SVM is best used in classification issues. The primary aim of SVM is to identify the best hyperplane that separates the data points of different classes in an N-dimensional space maximally[18]. It constructs a maximum-margin hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$.

Hyperparameter tuning is conducted using k -fold cross-validation.

E. Evaluation Metrics

To assess performance, several metrics are used:

Accuracy: The rate of correctly predicted features to all features. It gives an instant impression of how well the model is doing in general. The equation(9) shows how we get the accuracy of the obtained output.

Recall: The rate of true positive prediction features to all actual positive features. It is a measure of the proposed work's power to identify all cases that are important. It is shown in the equation(10).

F1-Score: F1 Score provides Harmonic mean of above mentioned precision and recall. It gives an overall measure of proposed work's performance, particularly helpful while working with class-imbalanced data. It is shown in the equation(11).

Precision: The true positive rate to predicted positives ratio. It is the number of the predicted positive instances that were indeed positive. It is shown in the equation(12).

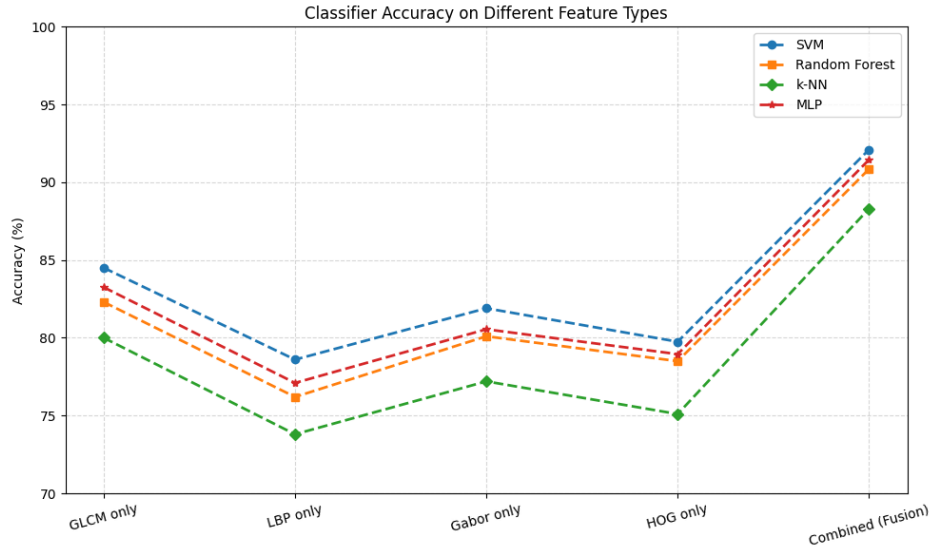


Fig 5: Comparison of accuracy for SVM,RF,k-NN and MLP classifiers

$$\text{Accuracy} = \frac{Ta1 + Ta2}{Ta1 + Ta2 + Fa1 + Fa2} \quad (9)$$

$$\text{Precision} = \frac{Ta1}{Ta1 + Fa1} \quad (10)$$

$$\text{Recall} = \frac{Ta1}{Ta1 + Fa2} \quad (11)$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

where Ta1 shows True Positive, Ta2 shows True Negative, Fa1 shows False Positive, and Fa2 shows False Negative. Confusion matrices and ROC-AUC scores are also employed to analyze misclassification and decision boundaries.

IV. RESULTS AND DISCUSSION

The present section presents the performance evaluation of the proposed textural feature extraction framework for rice seed variety. The results are evaluated in terms of classification accuracy, feature relevance, and classifier robustness.

The experiments were conducted on rice seed images comprising multiple varieties (e.g., Akatomachi, Koshihikari and Yangdao). The dataset includes 100–300 images per class, captured under controlled lighting. Each image underwent preprocessing and handcrafted feature extraction as detailed in the methodology.

The following classifiers were evaluated:

- Support Vector Machine (SVM) with RBF kernel
- k-Nearest Neighbors (k-NN)
- Multi-Layer Perceptron (MLP)
- Random Forest (RF)

A. Feature Evaluation and Classifier Performance

The features like GLCM, LBP,Gabor, HOG are classified and compared with different classifiers like SVM,RF,K-NN and MLP.It provides complete understanding of the results of the proposed method.

Table I summarizes the classification accuracy using individual and combined handcrafted features.

TABLE I: Classification Accuracy using Handcrafted Features

Feature Type	SVM (%)	RF (%)	k-NN (%)	MLP (%)
GLCM only	84.50	82.30	80.00	83.25
LBP only	78.60	76.20	73.80	77.10
Gabor only	81.90	80.10	77.20	80.55
HOG only	79.75	78.50	75.10	78.95
Combined (Fusion)	92.10	90.85	88.30	91.45

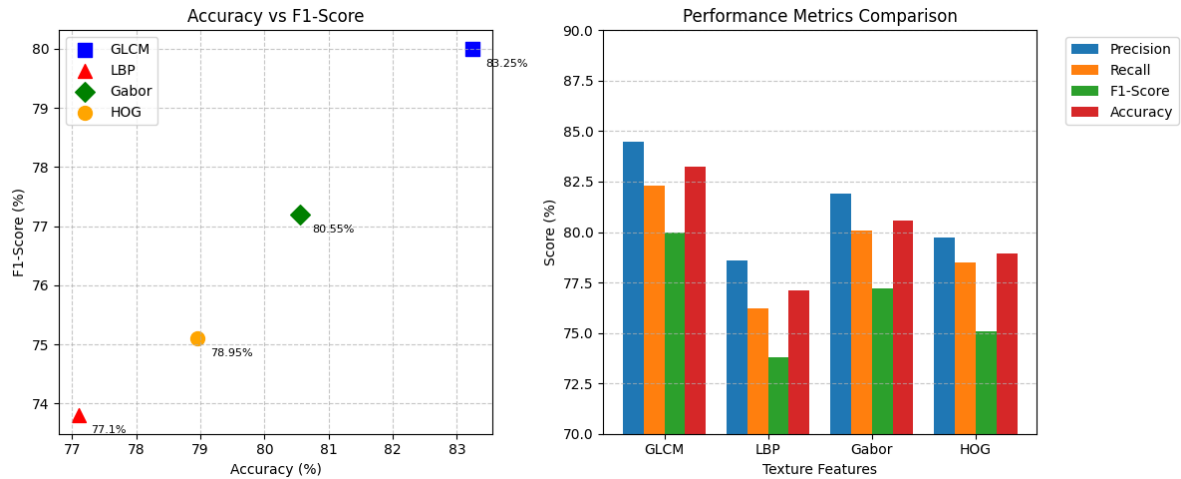


Fig 6: Comparison of Performance metrics with all the attributes

The results clearly demonstrate that the combination of handcrafted features leads to significantly higher classification accuracy compared to individual descriptors. This highlights the complementary nature of different textural attributes:

- **GLCM** captures local intensity variations and is effective for coarse textures.
- **LBP** encodes micro-textures and works well for fine-grain patterns.
- **Gabor filters** contribute orientation and frequency domain texture information.
- **HOG** captures the global shape and gradient structure of the seed boundary.

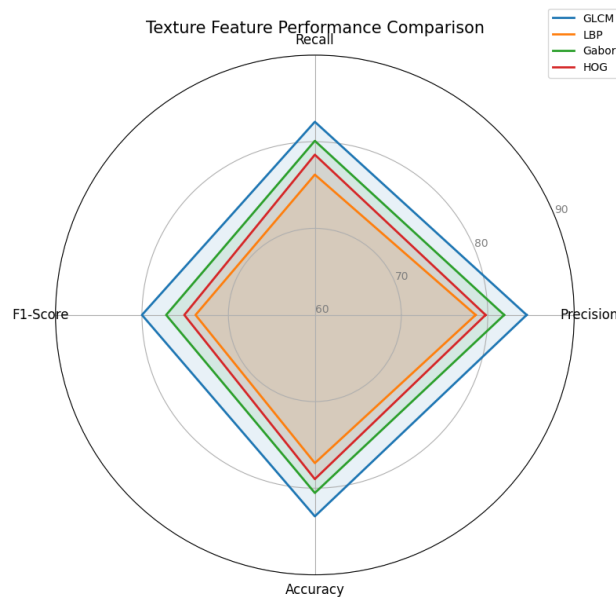


Fig 7: Performance Analysis of all features

TABLE II: Performance comparison of texture feature extraction methods (%)

Feature	Precision	Recall	F1-Score	Accuracy
GLCM	84.50	82.30	80.00	83.25
LBP	78.60	76.20	73.80	77.10
Gabor	81.90	80.10	77.20	80.55
HOG	79.75	78.50	75.10	78.95

Among classifiers, SVM outperforms others due to its ability to handle high-dimensional fused feature spaces with optimal margins. The random forest also shows competitive performance, highlighting the relevance of extracted features.

B. Summary of Findings

- Combined handcrafted features achieve over **92% accuracy**, meeting the goal of high classification performance.
- SVM is the most reliable classifier for this task.
- Textural descriptors are highly effective for rice seed variety classification without requiring deep learning-based feature extraction.

V. CONCLUSION

The textural analysis of rice seeds from different varieties using computational intelligence techniques has demonstrated significant potential in accurately classifying and characterizing rice grains based on their morphological and textural features. By leveraging **machine learning** algorithms and **image processing**, the study successfully differentiated between rice varieties with high precision, highlighting the robustness of computational methods in agricultural quality assessment. Key findings include the identification of distinct textural patterns (e.g., grain shape, surface roughness, and color distribution) that are critical for variety discrimination. Advanced **models such as convolutional neural networks (CNNs)** and texture-based feature extractors (e.g., GLCM, LBP) proved effective in capturing subtle differences among varieties, enabling automated, non-destructive classification. This approach not only enhances the efficiency of seed quality evaluation but also supports agricultural research, breeding programs, and food industry applications by providing rapid, scalable, and objective analysis. Future work could explore multi-modal data fusion (combining texture with spectral or genetic data) to further improve classification accuracy and expand the scope of computational intelligence in crop science.

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